FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – IV

JEE (Main)-2022

TEST DATE: 04-07-2022

ANSWERS, HINTS & SOLUTIONS

Physics **PART – A**

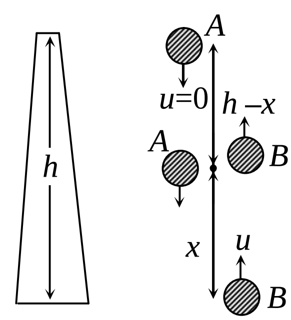
SECTION – A

1. B

Sol. If u is the initial speed of the second stone, then

$$0 = u^2 - 2g(4h)$$

or $u = \sqrt{8gh}$



If they meet at the height x from ground,

For A, $h - x = \frac{1}{2}gt^2$

For B, $x = (\sqrt{8gh})\, t - \frac{1}{2}gt^2$

$\therefore$ $h = \sqrt{8gh}\; t$ or $t = \sqrt{\frac{h}{8g}}$

2. C

Sol. At the top

$$T + mg = \frac{mv^2}{L}$$

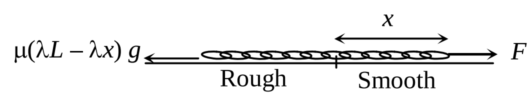
$$T < 10mg$$

$$v < \sqrt{11gL}$$

$$\sqrt{3gL} < u < \sqrt{13gL}$$

3. A

Sol. $F - \mu(\lambda L - \lambda x)g = \lambda L v \dfrac{dv}{dx}$



$$F\int_0^L dx - \mu\lambda g\int_0^L (L-x)dx = \lambda L\int_0^v v\,dv$$

$$FL - \mu\lambda g\left(L^2 - \frac{L^2}{2}\right) = \frac{\lambda L v^2}{2}$$

$$v = \sqrt{F - L}$$

4. A

Sol. $a \propto v^{1/3}$

$a = kv^{1/3}$

$$\frac{dv}{dt} = kv^{1/3}$$

$$\int_0^V \frac{dv}{v^{1/3}} = \int_0^t k\,dt\ ,\quad \frac{3}{2}v^{2/3} = t$$

$\therefore \quad v \propto t^{3/2}, \quad a \propto v^{1/3},\ a \propto t^{1/2}$

but, $\quad F \propto a$

$\therefore \quad F \propto t^{1/2}$

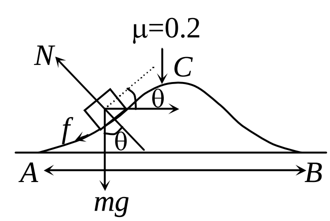
$P = Fv$

$P \propto t^{1/2} \times t^{3/2}$

$P \propto t^2$

5. C

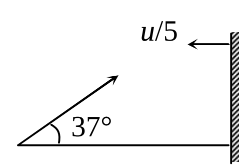
Sol. Work done by friction = $\int \vec{F}\cdot d\vec{s}$



$$= \int_0^x \mu mg\cos\theta \frac{dx}{\cos\theta} = \mu mgx = 20 \text{ J}$$

6. C

Sol. Let the ball collides with the wall after time t. Let velocity of ball after collision is v.



$$\frac{-v-\left(-\frac{u}{5}\right)}{-\frac{u}{5} - u\cos 37} = \frac{1}{4}, \quad -v + \frac{u}{5} = -\frac{u}{4}, \qquad v = \frac{u}{5} + \frac{u}{4} = \frac{9u}{20}$$

Also, $\quad (u\cos 37)t = \dfrac{9u}{20}(T - t)$

$$\frac{4ut}{5} = \frac{9u}{20}\left(\frac{2u}{g}\frac{3}{5} - t\right) \quad \Rightarrow \quad t = \frac{54u}{125g}$$

7. D

Sol. According to law of conservation of energy

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 \quad = \frac{1}{2}Mv^2 + \frac{1}{2}\times\frac{1}{2}MR^2\times\omega^2$$

or $$Mgh = \frac{1}{2}Mv^2 + \frac{1}{4}Mv^2 = \frac{3}{4}Mv^2, \qquad v = \sqrt{\frac{4}{3}gh}$$

Also $v = \sqrt{2ah}$

$\therefore$ $a = \frac{2}{3}g$

8. B

Sol. $v_e = \sqrt{\frac{2GM}{R}}$, $v = \frac{3}{4}v_e = \frac{3}{4}\sqrt{\frac{2GM}{R}}$

$$E_I = KE + PE = \frac{1}{2}mv^2 - \frac{GmM}{R} = \frac{1}{2}m\times\frac{9}{16}\times\frac{2GM}{R} - \frac{GmM}{R}$$

$$= \frac{9}{16}\frac{GmM}{R} - \frac{GmM}{R} = -\frac{7}{16}\frac{GmM}{R}$$

$$E_f = KE + PE = 0 - \frac{GmM}{r} = -\frac{GmM}{r} = \frac{GmM}{(R+h)}$$

$$-\frac{7}{16}\frac{GmM}{R} = -\frac{GmM}{(R+h)}$$

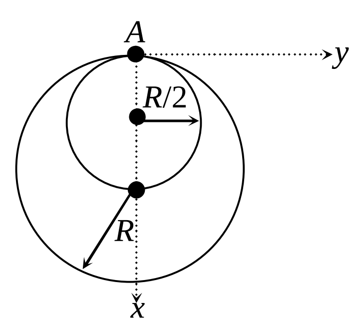
or $7(R+h) = 16R$ or $7h = 9R$ or $h = \frac{9R}{7}$.

9. A

Sol. $$\frac{1}{3}\left[A\left(\frac{h}{H}\right)^2\right]h\rho_w g = \frac{1}{3}AH\rho_w sg$$

$\Rightarrow$ $\frac{H}{h} = \frac{4}{3}$

10. C



Sol. $$x_{cm} = \frac{m\left(\frac{R}{2}\right) + 2mR}{m+2m} = \frac{5}{6}R$$

$$I_A = I_1 + I_2, \quad I_1 = m\left(\frac{R}{2}\right)^2 + m\left(\frac{R}{2}\right)^2 = \frac{mR^2}{2}$$

$$I_2 = 2mR^2 + 2mR^2 = 4mR^2 \Rightarrow I_A = \frac{9}{2}mR^2$$

for compound pendulum $T = 2\pi\sqrt{\frac{I}{Mgd}}$

$\Rightarrow$ here $M = 3m$, $d = \frac{5}{6}R \Rightarrow T = 2\pi\sqrt{\frac{9R}{5g}}$

11. B

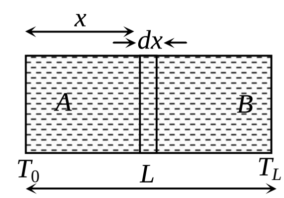
Sol. $f = \dfrac{(2n+1)v}{4(l+e)}$; $(l_1 + e) = \dfrac{v}{4f}$; $(l_2 + e) = \dfrac{3v}{4f}$ $\Rightarrow \dfrac{l_2 + e}{l_1 + e} = 3$

$l_2 = (3.6 - 2.34)$ m and $l_1 = (3.6 - 3.22)$ $\Rightarrow$ $e = 0.06$ m = 0.6 r $\Rightarrow r$ = 0.1 m

A = 100 π cm^2

12. A

Sol. $v = \sqrt{\dfrac{\gamma RT}{M}} = \sqrt{\dfrac{5RT}{3M}}$



$$dx = C \cdot dt = \sqrt{\frac{5R}{3M}\left[T_0 + \left(\frac{T_L - T_0}{L}\right)x\right]}\, dt$$

$$t = \frac{2L}{\sqrt{T_L} + \sqrt{T_0}}\sqrt{\frac{3M}{5R}}$$

13. C

Sol. $E_P = \left[\dfrac{\sigma}{2\varepsilon_0} + \dfrac{2\sigma}{2\varepsilon_0} + \dfrac{\sigma}{2\varepsilon_0}\right](-\hat{k}) = -\dfrac{2\sigma}{\varepsilon_0}\hat{k}$

14. A

Sol. $F_x = (1\times 10^{-6})(5-2x)\times 10^{6} = 5 - 2x$

$a_x = \dfrac{5-2x}{2}$, $v_x \dfrac{dv_x}{dx} = \dfrac{5-2x}{2}$

$$\int_0^0 2v_x\, dv_x = \int_0^{X_{\max}} (5-2x)dx \Rightarrow 5X_{\max} - X_{\max}^2 = 0$$

15. C

Sol. Applying KVL in loop ABCDA, ABFEA, ABHGA and ABJIA we get

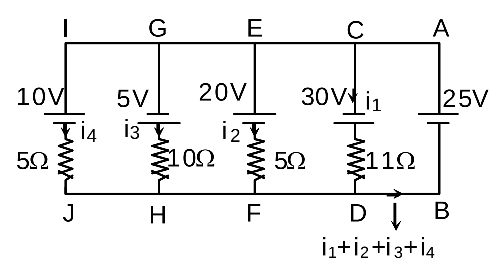
30 – i_1 × 11 = – 25 ..... (i)

20 + i_2 × 5 = 25..... (ii)

5 – i_3 × 10 = – 25 ..... (iii)

10 + i_4 × 5 = 25 ..... (iv)

i_1 = 5A, i_2 = 1A, i_3 = 3A and i_4 = 3A.

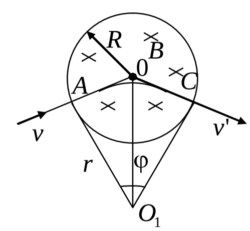


16. B

Sol. Magnetic force acting on charged particle is opposite to the direction of current. So the path will be helical with decreasing radius and because of magnetic force pitch will continuously increased.

17. B

Sol. The magnetic induction of the solenoid is directed along its axis. Therefore, the Lorentz force acting on the electron at any instant of time will lie in the plane perpendicular to the solenoid axis. Since the electron velocity at the initial moment is directed at right angles to the solenoid axis, the electron trajectory will lie in the plane perpendicular to the solenoid axis. The Lorentz force can be found from the formula $F = evB$.



The trajectory of the electron in the solenoid is an arc of the circle whose radius can be deter mined from the relation $evB = mv^2/r,$ whence

$$r = \frac{mv}{eB}$$

The trajectory of the electron in shown in figure, where O_1 is the centre of the arc AC described by the electron, v' is the velocity at which the electron leaves the solenoid. The segments OA and OC are tangents to the electron trajectory at points *A* and *C*. The angle between v and v' is obviously $\varphi = \angle AO_1C$ since $\angle OAO_1 = \angle OCO_1.$

In order to find $\varphi,$ let us consider the right triangle $OAO_1;$ side $OA = R$ and side $AO_1 = r.$

Therefore, $\tan(\varphi/2) = R/r = eBR/(mv).$

Therefore, $\varphi = 2\tan^{-1}\left(\frac{eBR}{mv}\right)$

Obviously, the magnitude of the velocity remains unchanged over the entire trajectory since the Lorentz force is perpendicular to the velocity at any instant. Therefore, the transit time of electron in the solenoid can be determined from the relation

$$t = \frac{r\varphi}{v} = \frac{m\varphi}{eB} = \frac{2m}{eB}\tan^{-1}\left(\frac{eBR}{mv}\right).$$

18. C

Sol. Magnetic flux through the loop A; $\phi = B\pi r^2 = \frac{\mu_0 I\ R^2}{2\left(R^2 + x^2\right)^{3/2}}\pi r^2$

Induced emf in the loop *A*; $\varepsilon = -\frac{d\phi}{dt} = \frac{\mu_0 I\ R^2 \pi r^2}{2}\left(-\frac{3}{2\left(R^2 + x^2\right)^{5/2}} 2x\frac{dx}{dt}\right)$

$$= -\frac{3\mu_0 IR^2 \pi r^2\ v}{2}\left[\frac{x}{\left(R^2 + x^2\right)^{5/2}}\right]$$

Induced emf is maximum when $\frac{d\varepsilon}{dx} = 0$

$\Rightarrow \quad \left(R^2 + x^2\right) - 5x^2 = 0$ or $x = \frac{R}{2}$

19. C

Sol. $\frac{1}{2}LI_0{}^2 = \frac{1}{2}(C_1 + C_2)V^2,\ V = \left[\frac{LI_0^2}{(C_1 + C_2)}\right]^{1/2},\ Q_1 = C_1V = C_1I_0\sqrt{\frac{L}{C_1 + C_2}}$

20. A

Sol. $\frac{2\pi}{\lambda}\left(\frac{yd}{D+v}\right) = 4\pi + \frac{\pi}{3};\ \frac{2\pi}{\lambda}\frac{D(3\lambda)}{D+v} = 4\pi + \frac{\pi}{3} \Rightarrow v = \frac{5}{13}$ m/s

SECTION – B

21. 00004.33

Sol. $m_1 = 1kg, m_2 = 2kg, u_1 = 3\,\text{ms}^{-1}, u_2 = 2\,\text{ms}^{-1}$

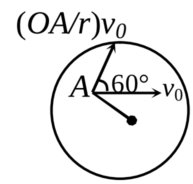
Initial momentum $= \sqrt{(m_1u_1)^2 + (m_2u_2)^2} = \sqrt{9+16}$ = 5 kg ms^{-1}

If combined velocity is v, $(m_1 + m_2)v = 5$, $v = \frac{5}{3}$ ms^{-1}

Heat liberated = loss in kinetic energy $= \frac{1}{2}m_1u_1^2 + \frac{1}{2}m_2u_2^2 - \frac{1}{2}(m_1+m_2)v^2 = \frac{13}{3}J$ =4.33J

22. 00004.36

Sol. $\left(\frac{OA}{R}\right)v_0 = \frac{2}{3}v_0$



$\therefore \quad v = \sqrt{v_0^2 + \frac{4}{9}v_0^2 + 2v_0\frac{2}{3}v_0\cos 60°}$

= 4.36 m/s

23. 00002.50

Sol. The height of water in the tank becomes maximum when the volume of water flowing into the tank per second becomes equal to the volume flowing out per sec. Volume of water flowing out per second is

$= Av = A\sqrt{2gh}$

Volume of water flowing in per second = 70 cm^3/sec

$A\sqrt{2gh} = 70, \quad 1\times\sqrt{2gh} = 70$ or $2\times 980\times h = 4900$

$\therefore \quad h = 2.5$ cm

24. 00001.41

Sol. Let the relation between T and $\frac{1}{V}$ be

$T = K\left(\frac{1}{V}\right)^n$

$\Rightarrow \quad 2T_0 = K\left(\frac{4}{V_0}\right)^n$ and $T_0 = K\left(\frac{1}{V_0}\right)^n$

$\Rightarrow \quad 2 = 4^n \Rightarrow \quad n = \frac{1}{2}$

Therefore $T = K\left(\frac{1}{V}\right)^{\frac{1}{2}}$ or $TV^{1/2} =$ constant

But for adiabatic process, $TV^{\gamma-1}$ = constant $\Rightarrow \gamma - 1 = \frac{1}{2}$ or $\gamma = \frac{3}{2}$

$\frac{V_{rms}}{V_{sound}} = \sqrt{\frac{3}{\gamma}} = \sqrt{\frac{3}{3/2}} = \sqrt{2}$

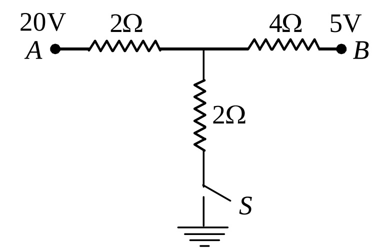
25. 00004.50

Sol. Let V be the potential of the junction as shown in the figure, applying junction law.

$40-2V+5-V=2V$

or V = 9Volt

$i_3=\frac{V}{2}=4.5$ Amp



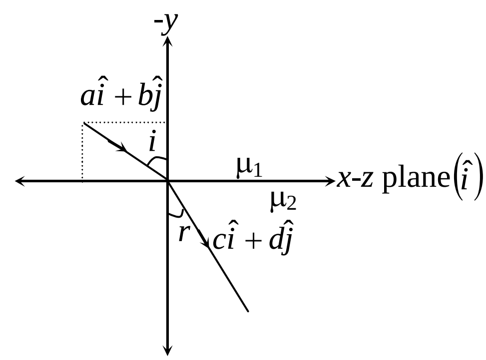
26. 00001.33

Sol. Applying Snell's law $\mu_1 \sin i=\mu_2 \sin r$

$$\left(\frac{3}{2}\right)\left(\frac{a}{\sqrt{a^2+b^2}}\right)=2\left(\frac{c}{\sqrt{c^2+d^2}}\right)$$

Here $\sqrt{a^2+b^2}=\sqrt{c^2+d^2}=1$

$\therefore \quad \frac{a}{c}=\frac{4}{3}$



27. 00002.39

Sol. $$E=C_1C_2\cos\omega_0 t+\frac{C_1C_3}{2}\cos(\omega_0+\omega)t+\frac{C_1C_3}{2}\cos(\omega_0-\omega)t$$

Of the three components, the highest frequency component will liberate the electrons with maximum kinetic energy

$(K.E)_{\max}+\phi=\frac{h}{2\pi}(\omega_0+\omega) \Rightarrow \phi$ = 2.39 eV.

28. 00000.27

Sol. $\frac{1}{\lambda_\alpha}=(Z-b)^2 R\left[\frac{1}{1^2}-\frac{1}{2^2}\right]$; $\quad \frac{1}{\lambda_\beta}=(Z-b)^2 R\left[\frac{1}{1^2}-\frac{1}{3^2}\right]$

$$\frac{\lambda_\beta}{\lambda_\alpha}=\frac{1-\frac{1}{4}}{1-\frac{1}{9}}=\frac{27}{32}$$

$\lambda_\beta=\frac{27}{32}\lambda_\alpha=\frac{27}{32}\times$ (0.32 Å) = 0.27 Å

29. 00001.97

Sol. $\sigma=e(n_e\mu_e+n_h\mu_h)$

$= 1.6\times10^{-19}(5\times10^{18}\times2.3+8\times10^{19}\times0.01)$

$= 1.97\ \Omega^{-1}m^{-1}$

30. 00002.63

Sol. p = 1mm, N = 100

Least count, $C = \frac{P}{N} = \frac{1\text{mm}}{100} = 0.01\text{mm}$

The instrument has a positive zero error $e = +NC = +4 \times 0.01 = +0.04\text{mm}$

Main scale reading is 2 × (1 mm) = 2 mm

Circular scale reading is 67 (0.01) = 0.67 mm

$\therefore$ observed reading is $R_0 = 2 + 0.67 = 2.67\text{ mm}$

So true reading = $R_0 - e$ = 2.63 mm

Chemistry

PART – B

SECTION – A

31. B
Sol. Factual

32. B
Sol. Lacks two alpha hydrogens

33. D
Sol. Among ketones, only methyl ketones give iodorm and bisulfite test

34. C
Sol. Markovnikov addition without rearrangement

35. B
Sol. Substitution, then addition

36. D
Sol. (A) $NH_2OH \rightarrow \overset{0}{N}_2$

$x + 3 - 2 = 0$

$x = -1$

$n = \text{factor} = 1 \times 1 = 1$

(B) n = factor of $Fe_2(SO_4)_3 \rightarrow 2$

Eq. weight = M/2

(C) Mili eq. = milimoles × n = factor

= 3 × 2

= 6.

37. A
Sol. For AgCl, molarity = Normality

Actual specific conductance = (specific conductance of AgCl –specific conductance of water)

$= (3.41 - 1.60) \times 10^{-6}$

$= 1.81 \times 10^{-6}\ \text{ohm}^{-1}\,\text{cm}^{-1}$

For saturated solution of sparingly soluble salt

$\Lambda_{eq} = \Lambda_{eq}^{\infty}$ and solubility = concentration

$$\Lambda_{eq}^{\infty} = \frac{1000 \times \text{specific conductance}}{\text{solubility}}$$

$$\therefore\ 138.3 = \frac{1000 \times 1.81 \times 10^{-6}}{S}$$

$$S(\text{mol L}^{-1}) = \frac{1000 \times 1.81 \times 10^{-6}}{138.3}$$

$= 1.31 \times 10^{-5}\ \text{mol L}^{-1}$

$AgCl \rightleftharpoons Ag^{+}(aq) + Cl^{-}(aq)$

$\therefore K_{sp} = [Ag^{+}][Cl^{-}]$

$S^2 = (1.31\times10^{-5})^2\,\text{mol}^2\text{L}^{-2}$
$= 1.72\times10^{-10}\,\text{mol}^2\text{L}^{-2}$

38. B
Sol. $HgI_2 + KI \rightarrow K_2[HgI_4]$
$\downarrow$
value of i decreases
$\Delta T_f = ik_f \times m$
$\Downarrow$
decreases
so T_f of the solution $\uparrow$

39. C
Sol. $\log\left(\dfrac{K_1 + \dfrac{11.11}{100}K_1}{K_1}\right) = \dfrac{E_a}{2.303R}\left[\dfrac{1}{300} - \dfrac{1}{301}\right]$

$E_a = \log(1.11) \times 2.303 \times 8.314 \times 300 \times 301)$
$= 78.362.8\text{J}$
$\log\left(\dfrac{K_2}{K_1}\right) = \dfrac{E_a}{2.303R}\left(\dfrac{1}{400} - \dfrac{1}{401}\right)$
$\log\left(\dfrac{K_2}{K_1}\right) = 0.0255$
$\dfrac{K_2}{K_1} = 1.06$
% increase $\approx 6.2\%$

40. A
Sol. Factual

41. D
Sol. Orbital angular momentum $= \sqrt{l(l+1)}\,\dfrac{h}{2\pi}$
For n = 5, maximum value of l = 4
Therefore, max orbital angular momentum $= \sqrt{4(4+1)}\,\dfrac{h}{2\pi}$

42. A
Sol. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
$K_p = P_{CO_2}$
According to Arrhenius equation:
$K = Ae^{-\frac{\Delta H}{RT}}$
$\log K_p = \log A - \dfrac{\Delta H_r^o}{2.303RT}$

$\log P_{CO_2} = \log A - \frac{\Delta H_r^o}{2.303}\frac{1}{RT}$ ...(1)

$Y = C + MX$

Graph (a) represent (i) and its slope can be used to determine the heat of the reaction.

43. D
Sol. All structures have same angle strain.

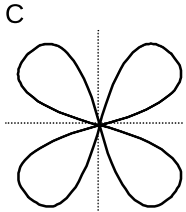
44. B
Sol. It is extraction of silver by leaching or cynide process.

45. C
Sol. $k = Ae^{-\frac{Ea}{RT}}$, as $T \to \infty, k \to A$

46. D
Sol. Here number of moles of reaction gases = number of moles of product gases.
Therefore, decreases in volume will not shift equilibrium, and concentration of both CO & CO_2 will increase, due to decrease in volume of reaction vessel.

47. B
Sol. This is cyclic hemiacetal of aldohexose

48. C
Sol.



$\pi^*(2p_x)$ Two nodal planes.

49. D
Sol. $\underset{4-x}{N_2O_5} \rightleftharpoons \underset{x-y}{N_2O_3} + \underset{x+y}{O_2}$

$\underset{x-y}{N_2O_3} \rightleftharpoons \underset{y}{N_2O} + \underset{y+x}{O_2}$

$\because$ $[O_2]$=x+y = 2.5

and 2.5 = $\frac{(x+y)(x-y)}{4-x}$

$\therefore$ $x = 2.166$

$[N_2O_5] = 4 - x = 1.846$

50. C
Sol. Factual

SECTION – B

51. 00010.50
Sol. $MgSO_4 + xH_2O \to MgSO_4.xH_2O$

$\Delta_r H_1 = -91.2 kj/mol$ ...(1)

$MgSO_4.7H_2O + (x-7)H_2O \to MgSO_4.xH_2O$

$\Delta_r H_2 = 13.8\,\text{kJ/mol} \quad ...(2)$

Eq. (2)-(1), $-105\,\text{kJ/mol} = \Delta x \qquad \frac{\Delta H}{10} = -10.5$

52. 00001.17

Sol. $\Delta T_f = k_f\left(\frac{n_1 + n_2}{W} \times 100\right)$

$= 1.86\left(\frac{\frac{3}{60} + \frac{1.8}{180} \times 1000}{95.2}\right) = 1.172$

$T_f = -1.172^{o}C$

53. 00068.35

Sol. Moles of Mg(g) = $\frac{1}{24} = 0.0417$

Energy required to convert Mg(g) to Mg^{+}(g) = $740 \times 0.0417 = 30.83\,\text{kJ}$

Remaining energy = 50-30.83 = 19.17 kJ

Moles of Mg^{2+}(g) formed $= \frac{19.17}{1450} = 0.0132$

Remaining Mg^{1+} = 0.0417 -0.132 = 0.0285

$\%Mg^{+} = \frac{0.285}{0.0417} \times 100 = 68.35\%$

54. 00000.38

Sol. $Cu^{2+} \xrightarrow[n_2=1]{G_2, E_2^0 = 0.15V} Cu^{+} \xrightarrow[n_1=1]{G_1, E_1} Cu$

$G_3, E_3^0 = 0.34V, n_3 = 2$

$G_1 + G_2 = G_3 \Rightarrow 1 \times E_1 + 1 \times 0.15 = 2 \times 0.34 \Rightarrow E_1 = 0.53V$

$Cu^{+} \quad + \quad Cu^{+} \quad \rightarrow \quad Cu^{2+} \quad + \quad Cu$

oxidation (anode)

$E_{RP}^{0} = 0.15V$

reduction(cathode)

$E_{RP}^{0} = 0.53V$

$E_{Cell}^{o} = 0.53 - 0.15 = 0.38V$

55. 00076.83

Sol. $B\% = \frac{(1.26 \times 10^{-4}) \times 100}{(1.26 \times 10^{-4}) + (3.8 \times 10^{-5})}$

$= 76.83\%$

56. 00022.00

Sol. Potato starch present in 10 mL of red gold sol for prevention of coagulation

$= \frac{0.44 \times 1000}{200} \times 10\text{mg} = 22\text{mg}$

Gold number of potato starch = 22.00

57. 00060.00

Sol. Isotonic solutions have equal osmotic pressures.

$\pi M_B = cRT$

$$c = \frac{8.2 \times 180}{0.082 \times 300} = 60\, gL^{-1}$$

58. 00006.61

Sol. The reaction occurring in the electrochemical cell are

At anode : $H_2 \longrightarrow 2H^+ + 2e^-$

At cathode : $\underline{Hg_2Cl_2 + 2e^- \longrightarrow 2Hg + 2Cl^-}$

Net cell reaction : $H_2 + Hg_2Cl_2 \longrightarrow 2H^+ + 2Hg + 2Cl^-$

$$\therefore E_{cell} = E^\circ_{Cl^-|Hg_2Cl_2|Hg} - E^\circ_{H^+|H^2} - \frac{0.059}{2}\log\frac{\left[Cl^-\right]^2\left[H^+\right]^2}{P_{H_2}}$$

$0.67 = 0.28 + 0.059\,pH$

$\therefore pH = 6.61$

59. 00004.27

Sol. $\frac{3K_1}{2K_2} = \frac{2}{3}$ $\qquad \frac{2K_2}{K_3} = \frac{3}{4}$

$\frac{K_1}{K_2} = \frac{4}{9}$ $\frac{K_2}{K_3} = \frac{3}{8} = \frac{9}{24}$

$K_1 : K_2 : K_3$

4 : 9 : 24

$$E = \frac{K_1E_1 + K_2E_2 + K_3E_3}{K_1 + K_2 + K_3}$$

$$= \frac{4x \times 8 + 9x \times 6 + 24x \times 3}{37x}$$

$$= \frac{32 + 54 + 72}{37} \approx 4$$

60. 00007.00

Sol. $n = 6$ to $n = 2$

	No. of lines
1st H – atom : $6 \to 5 \to 4 \to 3 \to 2$	4
2nd H – atom : $6 \to 4 \to 2$	2
3rd H – atom : $6 \to 3 \to 2$	1
	$\overline{\underline{7}}$

Mathematics

PART – C

SECTION – A

61. C

Sol. Let direction cosines of straight line be l, m, n $\therefore$ 4l + m + n = 0

$l - 2m + n = 0 \Rightarrow \frac{l}{3} = \frac{m}{-3} = \frac{n}{-9} \Rightarrow \frac{l}{-1} = \frac{m}{+1} = \frac{n}{3}$

$\therefore$ Equation of straight line is $\frac{x-2}{-1} = \frac{y+1}{1} = \frac{z+1}{3}$.

Hence (C) is the correct answer.

62. D

Sol. (a + c +e) + (b + d + f) = 1

So maximum value of (a + c + e) (b + d + f) is $\frac{1}{2} \times \frac{1}{2}$,

i.e.(ab + bc + cd + de+ ef) + (ad+ af + bc) $\le \frac{1}{4}$

or, ab + bc + cd+ de+ ef $\le \frac{1}{4}$ - (ad + af + bc) $\le \frac{1}{4}$

But if we take a = b = $\frac{1}{2}$ and c = d= e = f = 0, then equality holds, So maximum value is $\frac{1}{4}$

63. A

Sol. $\int_0^a f(x)dx = \frac{6a - \sin 2a}{4} \Rightarrow f(a) = \frac{3 - \cos 2a}{2} = 1 + \sin^2 a$

64. B

Sol. Let n= 2m +1

$A = \frac{1}{C_1} + \frac{1}{C_2} + \ldots + \frac{1}{C_m} = \frac{1}{C_{2m}} + \frac{1}{C_{2m-1}} + \ldots + \frac{1}{C_{m+1}}$

$\Rightarrow 2A + 2 = \sum_{r=0}^{n} \frac{1}{C_r}$

Let $S = \sum_{r=1}^{n} \frac{r}{C_r} = \sum_{r=0}^{n} \frac{r}{C_r} = \sum_{r=0}^{n} \frac{n-r}{C_{n-r}} = \sum_{r=0}^{n} \frac{n-r}{C_r}$.

$\Rightarrow 2S = n \sum_{r=0}^{n} \frac{1}{C_r} \Rightarrow S = n(A+1)$.

65. C

Sol. The chord of contact of tangents from (α,β) is

$\alpha x + \beta y = 1$ . . . . . (1)

Also, (α, β) lies on 2x +y = 4 , so that $2\alpha + \beta = 4$

$\Rightarrow \frac{\alpha}{2} + \frac{\beta}{4} = 1$

Hence, (1) passes through $\left(\frac{1}{2}, \frac{1}{4}\right)$.

66. A

Sol. Given expression = $i\left[\log\left(\dfrac{x-i}{x+i}\right)\right]-\pi+2\tan^{-1}x=k$ (say)

$\log\left(\dfrac{x+i}{x-i}\right)$ = $(k+\pi-2\tan^{-1}x)i$

or $\dfrac{x+i}{x-i}$ = $e^{i\theta}$ where $\theta = k+\pi-2\tan^{-1}x$

$\Rightarrow$ $(x+i) = (x\cos\theta+\sin\theta)+i\,(x\sin\theta-\cos\theta)$

$x = x\cos\theta+\sin\theta$ and $1 = x\sin\theta-\cos\theta$

$\Rightarrow$ $x = \cot(\theta/2) \Rightarrow \theta = 2\cot^{-1}x$

or $k+\pi-2\tan^{-1}x = 2\cot^{-1}x$

or $k+\pi = 2[\tan^{-1}x+\cot^{-1}x] = 2(\pi/2)$

or $k+\pi = \pi$

$k = 0$.

67. A

Sol. $Y-y = \dfrac{dy}{dx}(X-x)$

$\Rightarrow 5 = -\dfrac{1}{2}y^2\,\dfrac{dx}{dy}$

$10\dfrac{dy}{y^2} = -dx$

$\Rightarrow xy = 10$

68. C

Sol. $\Delta(0) = 0$ and $\Delta'(0) = 0$. so x^2 divides $\Delta(x)$.

69. A

Sol. $I(n) = \displaystyle\int_0^{\pi/2}\theta.\sin^n\theta\,d\theta$

$\Rightarrow$ $I(n) = \displaystyle\int_0^{\pi/2}\theta.\sin^{n-2}\theta\left(1-\cos^2\theta\right)d\theta$

$= I(n-2) - \displaystyle\int_0^{\pi/2}(\theta\,.\cos\theta\,).\cos\theta.\sin^{n-2}\theta\,d\theta$

$= I(n-2) - \theta.\cos\theta\,.\,\dfrac{\sin^{n-1}\theta}{n-1}\Bigg|_0^{\pi/2} + \displaystyle\int_0^{\pi/2}[\theta.(-\sin\theta)+\cos\theta].\dfrac{\sin^{n-1}\theta}{n-1}\,d\theta$

$= I(n-2) - \dfrac{1}{(n-1)}\displaystyle\int_0^{\pi/2}\theta.\sin^n\theta\;d\theta + \dfrac{1}{n-1}\displaystyle\int_0^{\pi/2}\cos\theta.\sin^{n-1}\theta\,d\theta$

$= I(n-2) - \dfrac{1}{(n-1)}.I(n) + \dfrac{1}{(n-1)(n)}.\sin^n\theta\,\Bigg|_0^{\pi/2}$

$\Rightarrow \dfrac{n}{n-1}I(n) = I(n-2) + \dfrac{1}{(n-1)(n)}$, $\Rightarrow I(n) - I(n-2)\,.\,\dfrac{n-1}{n} = \dfrac{1}{n^2}$.

70. B

Sol. The ellipse is $\frac{x^2}{4^2}+\frac{y^2}{5^2} = 1$

$\Rightarrow$ Parametric coordinates are $(4\cos\theta, 5\sin\theta)$

$\theta = \pi/2$, π and $3\pi/2$

$\Rightarrow$ Points are (0, 5), (– 4, 0) and (0, – 5)

$\Rightarrow$ Area of triangle = $\frac{1}{2}(4)(10)$ = 20 sq. units.

71. A

Sol. Given $f'(x) = \frac{-\sin^2 x+\frac{1}{2}}{f(x)}$

$\therefore\ f'(x).f(x) = \frac{-2\sin^2 x+1}{2}=\frac{\cos 2x}{2}$

$\therefore\ \int f'(x)\ f(x)dx=\frac{1}{2}\int\cos 2x dx$

or $\frac{[f(x)]^2}{2}+c=\frac{\sin 2x}{4}$

$f(x) = \pm\sqrt{\frac{1}{2}(\sin 2x)-2c}$

Which is a periodic function with period π.

Hence (A) is the correct answer

72. C

Sol. Let A and B be the points of observation with mid-point C. Let PQ be the flagstaff with mid-point R. The points A, B, P and Q are concyclic. If O is the centre of the circle through A, B, P, Q, then OR = $\frac{a+b}{2}$ and $\angle POR = \beta \Rightarrow$ PR = $\frac{a+b}{2}\tan\beta$

$\Rightarrow$ PQ = $(a + b)\tan\beta$

73. A

Sol. Let the tangent $\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$

$A \equiv (a\cos\theta, 0)$, $B \equiv (0, b\cot\theta)$

$\therefore\ P \equiv (a\cos\theta, b\cot\theta) = (h, k)$

$\sec\theta = \frac{a}{h}$, $\tan\theta = \frac{b}{k}$

$\therefore\ \frac{a^2}{h^2}=\frac{b^2}{k^2}+1$

$\therefore$ Locus is $\frac{a^2}{x^2}-\frac{b^2}{y^2}=1.$

74. A

Sol. Let $3x = \cos\theta \Rightarrow 3dx = -\sin\theta\, d\theta$

$$\therefore\ -\frac{1}{3}\int\frac{\frac{\cos\theta}{3}+\theta^2}{\sin\theta}\sin\theta d\theta=-\frac{1}{3}\int\left(\frac{1}{3}\cos\theta+\theta^2\right)d\theta$$

$$=-\frac{1}{9}\sin\theta-\frac{1}{9}\theta^3+c=-\frac{1}{9}\sqrt{1-9x^2}-\frac{1}{9}\left(\cos^{-1}3x\right)^3+c$$

Hence $A=-\frac{1}{9},\ B=-\frac{1}{9}$.

75. C

Sol. Since g(x) is continuous $\forall\ x\in R$, g(x) should be constant.

Since $f(x)\ \in(2,\sqrt{26})$,

$a\geq\sqrt{26}$, (as $\left[\frac{f(x)}{\sqrt{26}}\right]=0\ \forall\ x\in R$) .

So least integral value of a is 6.

76. B

Sol. Total number of permutations = $\frac{9!}{2!}$

Number of those containing 'HIN' = 7!

Number of those containing 'DUS' = $\frac{7!}{2!}$

Number of those containing 'TAN' = 7!

Number of those containing 'HIN' and 'DUS' = 5!

Number of those containing 'HIN' and 'TAN' = 5!

Number of those containing 'TAN' and 'DUS' = 5!

Number of those containing 'HIN', 'DUS' and 'TAN' = 3!

Required number = $\frac{9!}{2!}-\left(7!+7!+\frac{7!}{2}\right)+3\times5!-3!=169194$.

77. A

Sol. Let other end of diameter (h, k)

Hence centre is $\left(\frac{3+h}{2},\frac{k+4}{2}\right)$. This circle touches x-axis means $r=\frac{k+4}{2}$

$=\sqrt{\left(\frac{3+h}{2}-3\right)^2+\left(\frac{k+4}{2}-4\right)^2}$ gives the equation of parabola.

78. D

Sol. According to the given condition

${}^nC_3\left(\frac{1}{2}\right)^n={}^nC_4\left(\frac{1}{2}\right)^n$, where n is the number of times die is thrown

$\Rightarrow {}^nC_3={}^nC_4\Rightarrow n=7$.

Thus required probability $={}^7C_1\left(\frac{1}{2}\right)^7=\frac{7}{2^7}=\frac{7}{128}$.

79. A

Sol. We have $\frac{2}{\sqrt{c}+\sqrt{a}} = \frac{1}{\sqrt{b}+\sqrt{c}} + \frac{1}{\sqrt{a}+\sqrt{b}} = \frac{2\sqrt{b}+\sqrt{a}+\sqrt{c}}{\left(\sqrt{b}+\sqrt{c}\right)\left(\sqrt{a}+\sqrt{b}\right)}$

$\Rightarrow 2\sqrt{ab}+2b+2\sqrt{ac}+2\sqrt{bc} = 2\sqrt{bc}+2\sqrt{ac}+c+2\sqrt{ab}+a$

$\Rightarrow 2b = a + c$

The given numbers are in G.P. if

$2(bx + 1) = ax + 1 + cx + 1 \Rightarrow [2b - (a + c)]\, x = 0$

80. C

Sol. $x^2 + x - n = 0$, discriminent = $1 + 4n$ = odd number = D(say)

Now given equation would have a integral solution if D is a perfect square .

Let $D = (2\lambda + 1)^2 \Rightarrow n = \lambda + \lambda^2 = \lambda(\lambda + 1)$ = even number

$\Rightarrow$ n can be 2, 6, 12, 20, 30, 42, 56, 72, 90.

SECTION – B

81. 00000.28

Sol. $2\cos^{-1}x = \cos^{-1}(2x^2 - 1) \Rightarrow 2\cos^{-1}(4/5) = \cos^{-1}\left(\frac{32}{25} - 1\right)$

$\Rightarrow \cos(2\cos^{-1}(4/5)) = 7/25.$

82. 00008.00

Sol. $7! = 2^4 \times 3^2 \times 5 \times 7$

Since the factor should be odd as well as of the form 3t + 1, the factor cannot be a multiple of either 2 or 3. So the factors may be 1, 5, 7and 35 of which only 1 and 7 are of the from 3t +1, whose sum is 8.

83. 00001.00

Sol. Equation of normal to parabola is $y = mx - 4m - 2m^3$ (since a = 2)

it passes through (4, 2)

$2m^3 = -2 \Rightarrow m = -1$

$\therefore$ Only one normal is possible.

84. 00000.93

Sol. Probability of no tail in four throws

$= \frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{16}$

Probability of at least one tail = $1 - \frac{1}{16} = \frac{15}{16} = 0.9375$

85. 00050.00

Sol. We have $(x_1 + x_2 + x_3 + \ldots + x_{50})\left(\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_{50}}\right) \geq (50)^2$

[since A.M. $\geq$ H.M.]

$\Rightarrow \left(\frac{1}{x_1} + \frac{1}{x_2} + \ldots + \frac{1}{x_{50}}\right) \geq 50.$

86. 00090.00

Sol. Since $5x^2 + 12x + 13 = 0$ has imaginary roots as $D = 144 - 4 \times 5 \times 13 < 0$

So, both roots of $ax^2 + bx + c = 0$ and $5x^2 + 12x + 13 = 0$ will be common

$\therefore \frac{a}{5} = \frac{b}{12} = \frac{c}{13} \Rightarrow a^2 + b^2 = c^2 \Rightarrow \angle C = 90°$

87. 00002.00

Sol. The lines are $x + y = 1$, $x - y = 1$, $-x + y = 1$, $-x - y = 1$.
These lines form a square of side $\sqrt{2}$.
Area = $\sqrt{2}.\sqrt{2} = 2$.

88. 00000.20

Sol. $\cot^2 36° \cot^2 72° = (\cot^2 36° \cot^2 72° - 1) + 1$

$$= \left(\frac{\cos^2 36^0 \cos^2 72^0 - \sin^2 36^0 \sin^2 72^0}{\sin^2 36^0 \sin^2 72^0}\right) + 1$$

$$= \left(\frac{(1+\cos 144^0)(1+\cos 72^0) - (1-\cos 144^0)(1-\cos 72^0)}{4\sin^2 36^0 \sin^2 72^0}\right) + 1 = \frac{\cos 144^0 + \cos 72^0}{2\sin^2 36^0 \sin^2 72^0} + 1$$

$$= \frac{\cos 108^0 \cos 36^0}{\sin^2 36^0 \sin^2 72^0} + 1 = \frac{-\cos 72^0 \cos 36^0}{\sin^2 36^0 \sin^2 72^0} + 1 = 1 - \frac{(\cot 36^0 \cot 72^0)^2}{\cos 36^0 \cos 72^0}$$

$$= 1 - \frac{2\sin 36^0 (\cot 36^0 \cot 72^0)^2}{2\sin 36^0 \cos 36^0 \cos 72^0} = 1 - \frac{4\sin 36^0 (\cot 36^0 \cot 72^0)^2}{\sin 144^0}$$

$\Rightarrow \cot^2 36° \cot^2 72° = 1 - 4\cot^2 36° \cot^2 72°$

$\therefore 5 \cot^2 36° \cot^2 72° = 1$

$\Rightarrow \cot^2 36° \cot^2 72° = \frac{1}{5}$.

89. 00025.00

Sol. $\vec{a}.\vec{b} = 0 \Rightarrow x_1 + x_2 + x_3 = 0$

Thus we have to obtain the number of integral solution of this equation.

Coefficient of $x° \mid (x^{-3} + x^{-2} + x^{-1} + x^0 + x + x^2)^3$

$$= x° \left| \left(\frac{1 + x + x^2 + x^3 + x^4 + x^5}{x^3}\right)^3 \right.$$

$= x^9 \mid (1 - x^6)^3 (1 - x)^{-3} = {}^{11}C_9 - 3.{}^5C_3 = 25$.

90. 00120.00

Sol. Let θ be the angle between $\vec{a}$ and $\vec{b}$

$\vec{a} + \vec{b}$ will be a unit vector if and only if $(\vec{a} + \vec{b})(\vec{a} + \vec{b}) = 1$

i.e. $\vec{a}.\vec{b} = -\frac{1}{2}|\vec{a}||\vec{b}|$ $\cos\theta = -\frac{1}{2}$ or $\theta = 120$